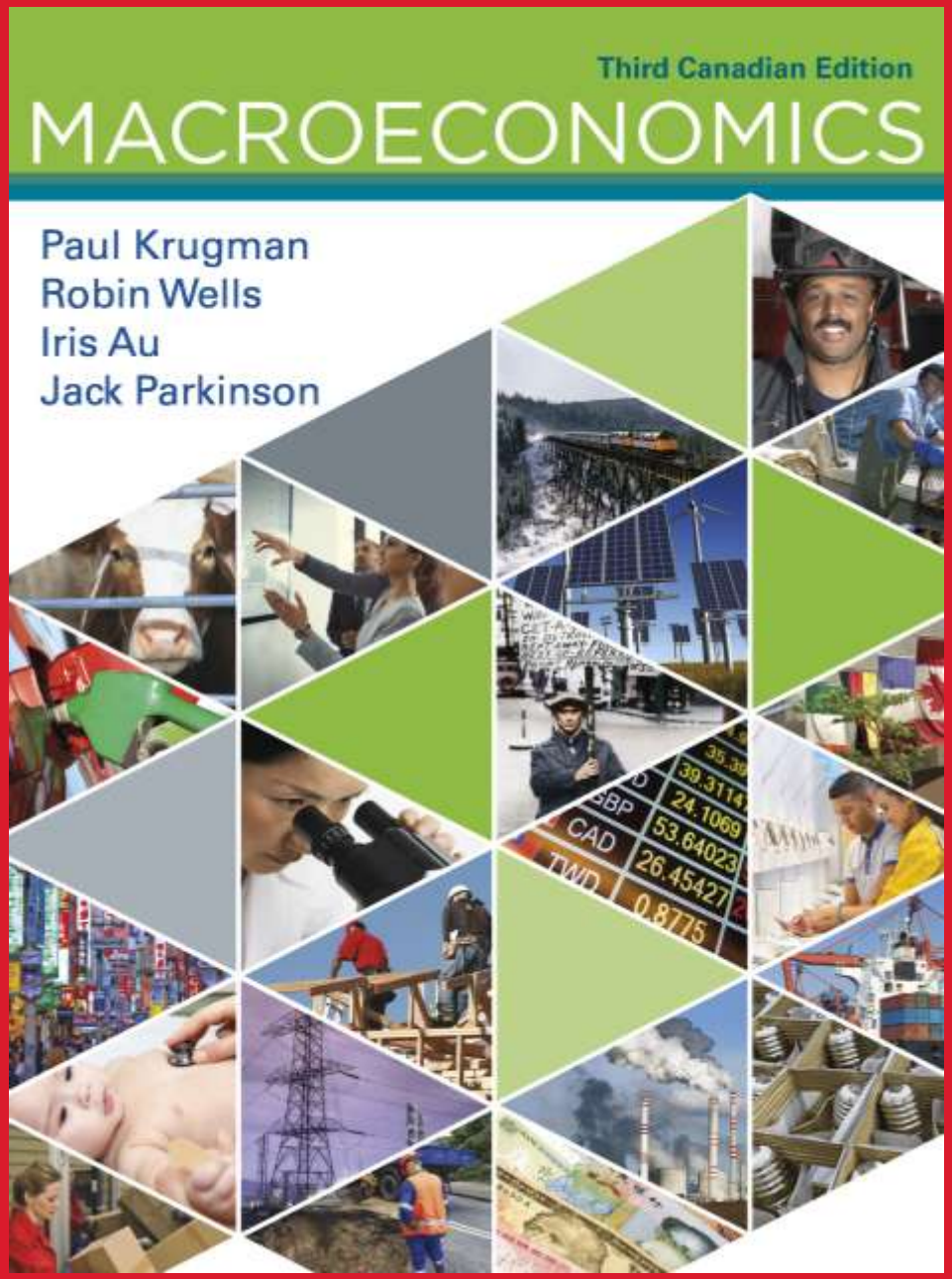
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7

# GDP AND THE CPI: TRACKING THE MACROECONOMY

# WHAT YOU WILL LEARN IN THIS CHAPTER

- How do economists use aggregate measures to track the performance of the economy?
- What is gross domestic product (GDP) and how is it calculated?
- What is the difference between real GDP and nominal GDP and why is real GDP the appropriate measure of real economic activity?
- What is a price index and how is it used to calculate the inflation rate?

# HOW DO YOU MEASURE AN ECONOMY?

- Is China's economy bigger than the Canada economy?
- **How can you compare the sizes of two economies when they produce different things?**

# AN EXPANDED CIRCULAR FLOW DIAGRAM

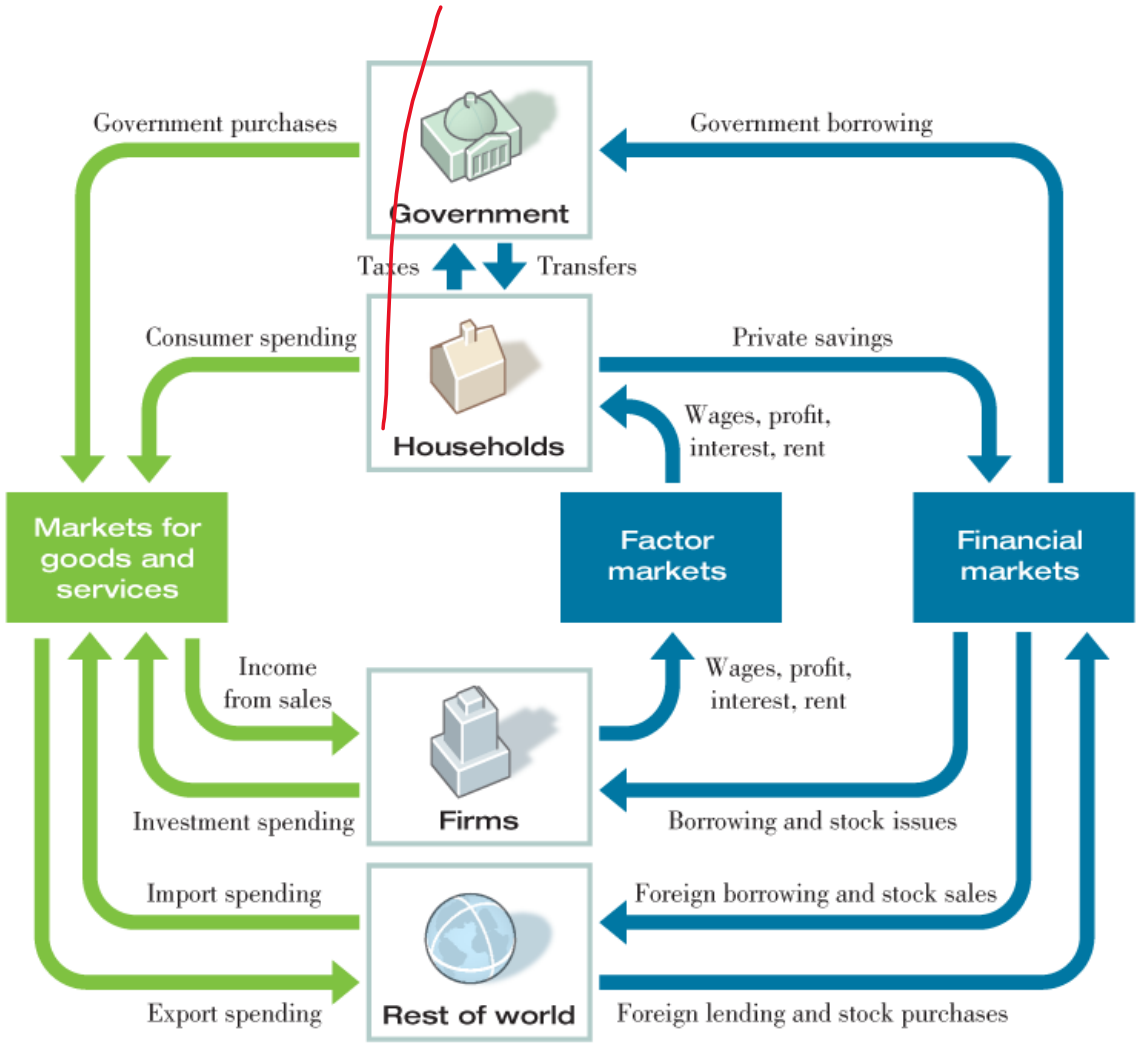



**FIGURE 7-1** Krugman/Wells/Au/Parkinson, *Macroeconomics,* 3e, Canadian, © 2018 Worth Publishers

# THE NATIONAL ACCOUNTS (Tracks where the spending is going)

- **Consumer spending**
- **Government purchases of goods and services**
- **Investment spending (Spending on physical capital and changes to inventories)**
- **Exports and imports**

# WHAT IS GDP?

- **Gross domestic product (GDP):**

The market value of all final goods and services produced within a country in a year

- **GDP per capita** = GDP / population
- GDP per capita finds the average wealth of an individual

# Avoiding Multiple Counting

- To avoid multiple counting, only goods and services are counted. goods are not counted.

- **Final goods : Goods/Srevices purchased for final use**

**Ex: Cars**

- **Intermediate goods: For resale or manufacture**

Ex: Tires, Steel

- **Value added For a producer is the value of sales – the value of intermediate goods**

# GDP—WHAT'S IN AND WHAT'S OUT

- **Included**
  - Domestically produced final goods and services, capital goods, new construction of structures, and changes to inventories
  - Capital goods
  - New construction of structures
  - Changes to inventories
- **Not Included**
  - …… goods and services
  - Inputs
  - ..
  - Financial assets
  - G&S produced ____ this country

# GDP—WHAT'S IN AND WHAT'S OUT

- Not included
  - Intermediate goods and services
  - Inputs
  - Used goods
  - Financial assets
  - Goods and Services produced abroad
  - Volunteered work, underground/illegal activities (selling meth is not included in GDP)
  - Household (HH) production (things like cooking, caring for the child, etc.)
  - Environmental damage

# CALCULATING GDP

- 3 methods:

1. The Value-Added Approach: Add up the total **value of all final goods and services** produced
2. The Expenditure Approach: Add up all **spending** on domestically produced final goods and services
3. The Income Approach: Add up factor income earned from firms

# CALCULATING GDP: Example

*2. Aggregate expenditure on domestically produced final goods and services = $21,500*

| | Canadian Ore, Inc. | Canadian Steel, Inc. | Canadian Motors, Inc. | Total factor income |
|---|---|---|---|---|
| **Value of sales** | $4,200 (ore) | $9,000 (steel) | $21,500 (car) | |
| **Intermediate goods** | 0 | 4,200 (iron ore) | 9,000 (steel) | |
| **Wages** | 2,000 | 3,700 | 10,000 | $15,700 |
| **Interest payments** | 1,000 | 600 | 1,000 | 2,600 |
| **Rent** | 200 | 300 | 500 | 1,000 |
| **Profit** | 1,000 | 200 | 1,000 | 2,200 |
| **Total expenditure by firm** | 4,200 | 9,000 | 21,500 | |
| **Value added per firm = Value of sales – Cost of intermediate goods** | 4,200 | 4,800 | 12,500 | |

*3. Total payments to factors = $21,500*

*1. Value of production of final goods and services, sum of value added = $21,500*

**FIGURE 7-2** Krugman/Wells/Au/Parkinson, *Macroeconomics*, 3e, Canadian,
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# METHOD 2: Expenditure Approach

Add up all spending on domestically produced final goods and services.

This results in the equation GDP = C+I+G+X – IM

*I = Investment (from firms)*

*C = Consumption (household/consumer)*

*G = Govt. Spending*

***X-IM is net exports***

# METHOD 3: Income Approach

Add up factor income earned from firms.

**Factor Income:** earned by factors of production. (Wage, interest, rent, proft)

**Non-factor payments** earned by government as a result of the production.

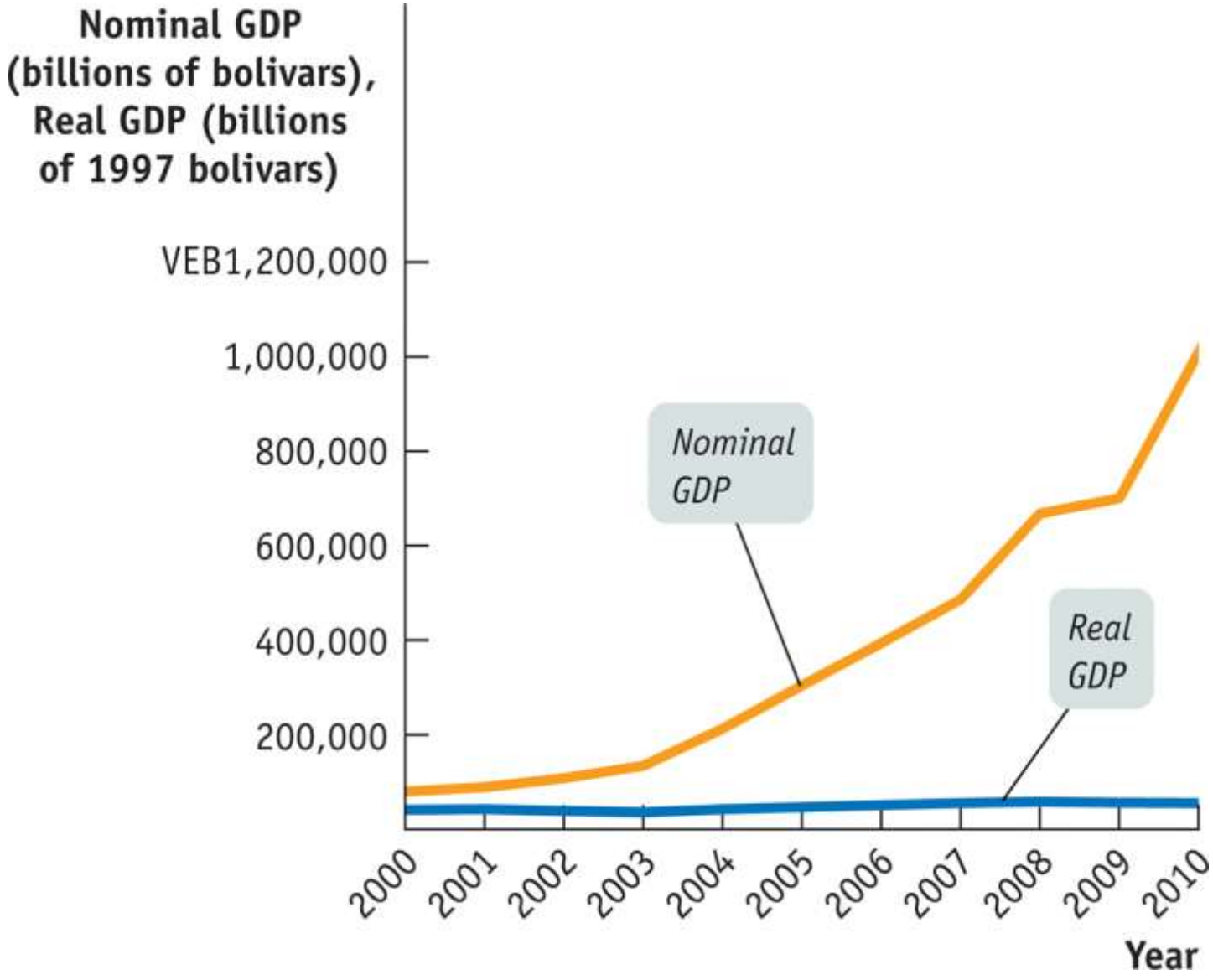
Net taxes = taxes – subsidies

Capital Depreciation

# REAL GDP: A MEASURE OF AGGREGATE OUTPUT

- We need to be able track the quantity of total output over time.
- **Real GDP**: calculated using the prices of a base year
- **Nominal GDP**: calculated using current prices

# How Venezuela is Doing?



# REAL VS NOMINAL GDP

## TABLE 7-1 Calculating GDP and Real GPD in a Simple Economy

| | In Year 1 Quantity (billions) | In Year 1 Price | In Year 2 Quantity (billions) | In Year 2 Price |
|---|---|---|---|---|
| Apples | 2000 | $0.25 | 2200 | $0.30 |
| Oranges | 1000 | $0.50 | 1200 | $0.70 |
| GDP (billions of dollars) | (2000 × $0.25) + (1000 × $0.50) = **$1000** | | (2200 x 0.30) + (1200 x 0.70) = 1500 | |
| Real GDP (billions of year 1 dollars) | | | | |

Base year = year 1

# REAL vs Nominal GDP

**TABLE 7-2 Nominal versus Real GDP in 2003, 2009, and 2015**

| | Nominal GDP (billions of current dollars) | Real GDP (billions of 2007 chained dollars) |
|---|---|---|
| 2000 | $1,102.4 | $1,323.2 |
| 2007 | 1,573.5 | 1,573.5 |
| 2016 | 2,027.5 | 1,796.2 |

# PRICE INDEXES AND THE AGGREGATE PRICE LEVEL

- To measure the aggregate price level, economists calculate the cost of purchasing a **market basket.**
- **Market basket:** a hypothetical set consumer goods and services that they might need

# MARKET BASKETS AND PRICE INDEXES

## TABLE 7-3 Calculating the Cost of a Market Basket

| | Pre-frost | Post-frost |
|---|---|---|
| Price of apple | $0.20 | $0.40 |
| Price of peach | 0.60 | 1.00 |
| Price of small bunch of grapes | 0.25 | 0.45 |
| Cost of market basket (200 apples, 50 peaches, 100 small bunch of grapes) | (200 × $0.20) + (50 × $0.60) + (100 × $0.25) = $95.00 | (200 × $0.40) + (50 × $1.00) + (100 × $0.45) = $175.00 |

# PRICE INDEXES

- **Price index in a given year:**

$$\frac{Cost\ of\ market\ basket\ in\ a\ given\ year}{Cost\ of\ market\ basket\ in\ base\ year} * 100$$

- The CPI (consumer price index) measures the cost of the market basket in a typical urban family

# INFLATION RATE, CPI, AND OTHER INDEXES

- The **inflation rate:** the yearly percentage change in a price index such as CPI.
- Inflation rate:

$$\frac{PriceIndexYear2 - PriceIndexYear1}{PriceIndexYear1} * 100$$

# OTHER PRICE MEASURES

- **Producer price index (PPI):** measures changes in the prices of goods purchased by producers

- **GDP deflator** calculated as the ratio of nominal to real gdp

$$\text{GDP Deflator} = \frac{Nominal\ GDP\ for\ given\ year}{Real\ GDP\ for\ given\ year} * 100$$

# THE CPI, THE PPI, AND THE GDP DEFLATOR

The three measures usually move closely together.

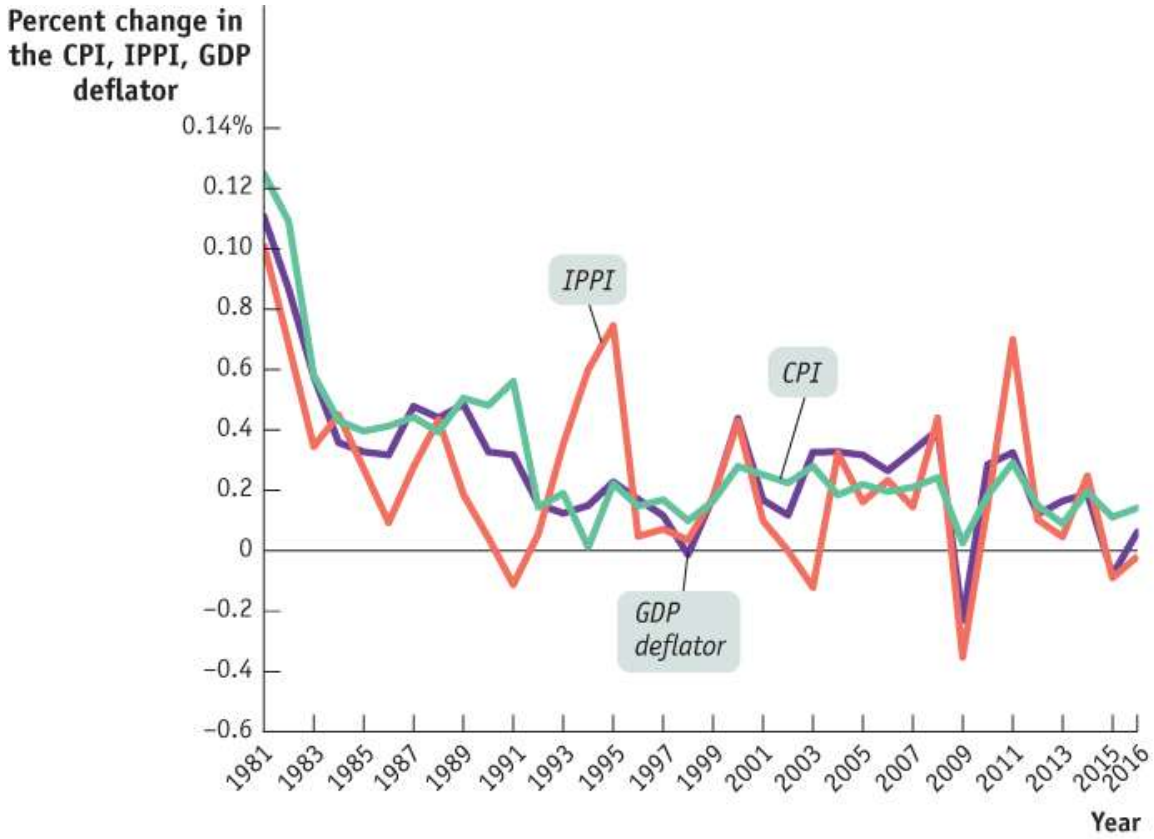



**FIGURE 7-6** Krugman/Wells/Au/Parkinson, *Macroeconomics,* 3e, Canadian, © 2018 Worth Publishers
*Data from:* Statistics Canada